INCH-POUND

BB-H-1168D
15 November 2010
SUPERSEDING
BB-H-1168C
3 August 2000

FEDERAL SPECIFICATION

HELIUM, TECHNICAL

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two technical grades of gaseous helium (see 6.1).

1.2 Classification. The helium shall be of the following grades, as specified (see 6.2):

Grade A - 99.995 percent helium
Grade B - 97.5 percent helium

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification

RR-C-901 - Cylinders, Compressed Gas: Seamless Shatterproof, High Pressure DOT 3AA Steel, and 3AL Aluminum

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Logistics Agency Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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Federal Standard

FED-STD-123	- Marking for Shipment (Civil Agencies)

Commercial Item Description

A-A-59860	- Valves, Cylinder, Gas (for Compressed or Liquefied Gases)

(Activities outside the federal government may obtain copies of federal specifications, standards, and commercial item descriptions as specified in the General Information section of the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index is for sale on a subscription basis from the General Services Administration, Federal Supply Service, Specification Section, East 470 L'Enfant Plaza SW, Suite 8100, Washington, DC 20407.)

(Single copies of this specification, and other federal specifications, standards and commercial item descriptions required by activities outside the federal government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service, Specification Section, East 470 L'Enfant Plaza SW, Suite 8100, Washington, DC 20407.)

(Federal government activities may obtain copies of federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies. Electronic copies may be obtained from https://assist.daps.dla.mil/.)

Military Standards

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders
MIL-STD-129	- Military Marking for Shipment and Storage
MIL-STD-1411	- Inspection and Maintenance of Compressed Gas Cylinders

(Copies of military standards required by contractors in connection with specific procurement functions are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. Electronic copies may be obtained from https://assist.daps.dla.mil/.)

Federal Regulations

43 CFR 3195	- Helium Contracts
49 CFR 171-181	- Hazardous Materials Regulations

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(The Code of Federal Regulations (CFR) is for sale on a subscription basis from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuing them. Electronic copies may be obtained from http://www.gpoaccess.gov/ecfr/.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Bureau of Land Management (BLM)

Information Circular (IC) 8367	-	Computing Volume of Helium in Cylindrical Steel Containers at 10 to 10,000 psia
Report of Investigations (RI) 7444	-	An Improved Method and Apparatus for Analysis of Impurities in Helium

(The above circular and report were drafted by the Bureau of Mines (USBM), which has been absorbed by the BLM. Application for copies should be addressed to the Bureau of Land Management, Helium Operations Office, 801 South Fillmore Street, Suite 500, Amarillo, TX 79101-3545.)

Compressed Gas Association (CGA)

CGA P-15	-	Filling of Industrial and Medical Nonflammable Compressed Gas Cylinders

(Application for copies should be addressed to the Compressed Gas Association, 4221 Walney Road, 5th Floor, Chantilly, VA 20151-2923. Electronic copies may be obtained from http://www.cganet.com/.)

3. REQUIREMENTS

3.1 Material. Helium shall be Bureau of Land Management helium pursuant to 43 CFR 3195 (see 6.3).

3.2 Pressure. Each container shall be filled to the maximum pressure authorized by 49 CFR 171-181, when tested as specified in 4.5.1. The net content of helium shall be corrected to 70 °F and 14.7 pounds per square inch absolute (psia) and reported in standard cubic feet as shown in IC 8367.

3.3 Leakage. Each container shall show no evidence of leakage when tested as specified in 4.5.2.

3.4 Helium (grade A). Grade A helium shall contain not greater then 50 parts per million (ppm) by volume aggregate impurities, when tested as specified in 4.5.3.

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3.5 Helium (grade B). Grade B helium shall exhibit non-flammability and buoyancy, when tested as specified in 4.5.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Points of inspection. When helium containers are filled from production plant systems or other storage facilities for delivery to destination by any mode of transportation, quality conformance inspection shall be conducted on the containers and their contents prior to leaving the site of filling or shipment.

4.3 Examination of preparation for delivery. The packing and marking of the cylinders shall be examined to determine compliance with the requirements of section 5.

4.3.1 Individual tests. Each container shall be tested for filling pressure and leakage as specified in 4.5.1 and 4.5.2.

4.3.2 Sampling tests. When applicable, helium shall be sampled and tested, as specified in 4.5.3 or 4.5.4.

4.3.2.1 Lots. When helium cylinders are filled, each group of cylinders charged on the same manifold at the same time shall constitute a lot. When helium tanks, as part of a tank car or tube trailer, are filled, each tank shall constitute a lot, unless all tanks are part of a common manifold.

4.3.2.2 Sampling plan. The sample unit for testing shall be one filled sample container, containing at least one cubic foot of gas. The sample size for each cylinder lot shall conform to table I. In addition, with the exception of its initial service, each tank of each tank car or tube trailer shall be sampled prior to filling and after approximately one cubic foot of gas has been purged. It may be necessary to add gas to the tank in order to perform this test. When the tanks of a tank car or tube trailer are part of a common manifold, one sample will be sufficient. The lot shall be rejected if a sample fails to pass any test.

TABLE I. Sampling size for cylinder lots.

Lot size	Sample size
1 to 10 cylinders	1
11 to 40 cylinders	2
41 to 70 cylinders	3
71 or more cylinders	4

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4.4 Testing. Testing of the helium shall be classified as individual tests or sampling tests as specified herein.

4.5 Test methods.

4.5.1 Pressure. Measure pressure to the nearest 5 pounds per square inch gauge (psig) by attaching a pressure gauge having minor scale gradations of 10 psig or less to the valve outlet and attaching a thermocouple or mercury-in-glass thermometer having scale divisions of 1 °F to the container wall. If a thermometer is used, apply tape or putty to the bulb to insulate it from ambient temperatures, but do not apply it between the bulb and the container wall. For service pressures of 1800-2640 psig, use the Pressure-Temperature Conversion Chart as specified in CGA P-15. Open the container valve and observe the internal gauge pressure. The internal pressure should not vary from the maximum marked nominal pressure by more than 25 psig.

4.5.2 Leakage. Test for leakage by brushing some leak-detecting solution over the valve and the junction between the valve and the container, but be careful not to get any solution on the valve outlet. Test for leakage through the closed valve outlet by connecting one end of a tube over the valve outlet and immersing the other end in a container of liquid. Any evidence of gas bubbling through the liquid shall be considered evidence of leakage.

4.5.3 Helium (grade A). Determine percent by volume (v/v) for grade A helium with a thermal conductivity analyzer by measuring the aggregate impurities, which have a different thermal conductivity than helium. Calibrate the analyzer to appropriate intervals with calibration gas standards. The range of the analyzer should be no greater than ten times the difference between the specified percent helium and 100 percent. Thus for 99.995 percent helium the analyzer should have a maximum range of 500 ppm impurity.

(Standards used for calibration shall be Bureau of Land Management certified standards, obtained from the bureau of the address in 6.3. Alternatively, standards may be prepared and the synthetic mixture analyzed in accordance with RI 7444.)

4.5.4 Helium (grade B).

4.5.4.1 Non-flammability. Test for non-flammability of grade B helium by collecting a sample in an inverted test tube and inserting a burning splinter of wood up into the tube. (Note: Use caution.) The flame should be extinguished without an audible pop.

4.5.4.2 Buoyancy. Test for buoyancy of grade B helium by collecting a sample in a toy balloon, tying the open end and releasing the balloon. The balloon should show buoyancy.

5. PACKAGING

5.1 Packing. Cylinders shall conform to, and shall be packed in accordance with 49 CFR 178. In addition, helium shall be provided in government-furnished cylinders conforming to RR-C-901. When specified (see 6.2), helium shall be provided in contractor-furnished cylinders conforming to RR-C-901, equipped with valves conforming to A-A-59860. When specified

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(see 6.2), helium shall be furnished in contractor-owned cylinders. All cylinders shall be inspected and maintained in accordance with MIL-STD-1411 for helium cylinders.

5.2 Marking. Marking shall be for civil agencies or military activities, as specified (see 6.2).

5.2.1 Civil agencies. Marking for shipment to civil agencies shall be in accordance with FED-STD-123.

5.2.2 Military activities. Marking for shipment to military activities shall be in accordance with MIL-STD-129. In addition, cylinders shall be tagged and color-coded as specified below.

5.2.2.1 Identification tag. An identification tag impervious to climatic conditions shall be wired to the outlet port of each container and shall bear the following information: proper shipping name, specification number with revision letter, grade, NSN, quantity, name of manufacturer, name of contractor (if different from manufacturer), date of manufacture, lot identification number, and lot analysis.

5.2.2.2 Color codes. Department of Defense (DoD)-owned cylinders shall be color-coded in accordance with MIL-STD-101.

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Grade A helium is intended for use as a shielding gas in arc welding. Grade B helium is intended for use as a buoyant in lighter-than-air craft such as weather balloons.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Grade required (see 1.2 and 6.1).
c. Packaging requirements (see 5.1).
d. When other than government-furnished cylinders are to be used (see 5.1).
e. Marking required (see 5.2).
f. When procurement shall be made on a cost-per-unit bases in accordance with MIL-STD-1411.

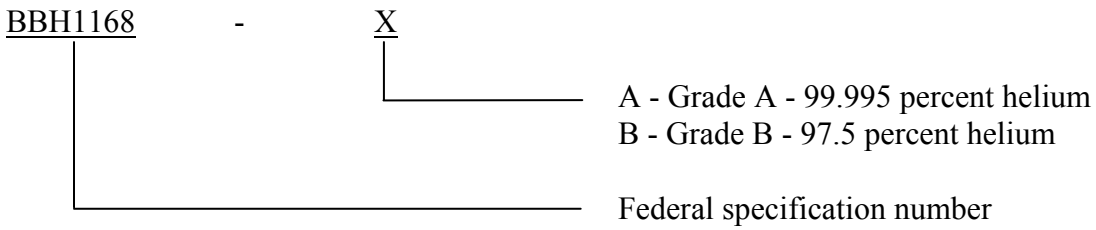
6.3 Bureau of Land Management helium. In order to supply helium to a Federal agency and its contractors, a private helium distributor is required to be an eligible supplier of Bureau of Land Management helium. A private distributor who desires to establish its eligibility may contact the Bureau of Land Management at the following address:

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Bureau of Land Management
Helium Operations Office
801 South Fillmore Street, Suite 500
Amarillo, TX 79101-3545

The Bureau of Land Management will also maintain at this address a current list of eligible private helium distributors. Upon request, a copy of this list will supplied to Federal agencies, their contractors, and any other interested persons.

6.4 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



6.5 Subject term (key word) listing.

Analyzer, thermal conductivity
BLM
Buoyancy
Bureau of Land Management
Bureau of Mines
Color code
Cylinder
Identification tag
Impurities
Leakage
Marking
Non-flammability
Percent by volume
Pressure
Pressure-temperature conversion
Sampling
USBM
Valve

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

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MILITARY INTERESTS:

Custodians:
Navy - SH
Air Force - 68
DLA - GS

Review Activities:
Air Force - 19
Other - DS

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS
HHS - NIH
NA - OCE
Interior - BLM
DOT - ACO

Preparing Activity:
DLA - GS3

(Project 6830-2010-003)

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